R, S partial orders on A $\qquad R = \{(x,y) \dots\}$ $\quad S = \{(a,b) \dots\}$

$R \cap S$, $R \cup S$ are partial orders on A?

reflexive: $\forall x \in A : (x,x) \in R \cap S$

$$\left.\begin{array}{l} R \text{ is reflexive} \Rightarrow (x,x) \in R \\ S \quad " \quad " \Rightarrow (x,x) \in S \end{array}\right\} \Rightarrow (x,x) \in R \cap S \quad \checkmark$$

transitive: $\forall_{x,y,z} : (x,y) \in R \cap S \wedge (y,z) \in R \cap S \Rightarrow (x,z) \in R \cap S$

$$\left.\begin{array}{l} (x,y) \in R \wedge (y,z) \in R \xrightarrow[\text{trans.}]{R} (x,z) \in R \\ (x,y) \in S \wedge (y,z) \in S \xrightarrow[\text{trans.}]{S} (x,z) \in S \end{array}\right\} \Rightarrow (x,z) \in R \cap S \quad \checkmark$$

antisymmetric: $\forall x,y \in A : (x,y) \in R \cap S \wedge (y,x) \in R \cap S \Rightarrow x = y$

First proof:

$$(x,y) \in R \wedge (y,x) \in R \xrightarrow[\text{antisymmetric}]{R} x = y \quad \checkmark$$

Second proof: (Contradiction)

assume $R \cap S$ is not antisymmetric

thus there are $x,y \in A$ s.t. $(x,y) \in R \cap S \wedge (y,x) \in R \cap S$ but $x \neq y$

$$\underbrace{(x,y) \in R \quad \wedge \quad (y,x) \in R}$$

Since R is antisymmetric $x = y$. this contradicts.

Inverse of a relation R on A is a relation R^{-1} on A defined as

$$R^{-1} = \{(b,a) : (a,b) \in R\}$$

$R: a \longrightarrow b$ $\qquad$ $R^{-1}: a \longleftarrow b$

exmp: $A = \{2,3,4\}$

$R = \{(a,b) : a \mid b\}$

$R = \{(2,2), (2,4), (3,3), (4,4)\}$

$R^{-1} = \{(2,2), (4,2), (3,3), (4,4)\}$

Prove that if (R, A) is poset then (R^{-1}, A) is also a poset.

- R^{-1} is reflexive H.W.
- R^{-1} is antisymmetric H.W.
- R^{-1} is transitive: $\forall_{x,y,z}: \; x R^{-1} y \wedge y R^{-1} z \Rightarrow x R^{-1} z$

$$(x,y) \in R^{-1} \wedge (y,z) \in R^{-1} \Rightarrow (x,z) \in R^{-1}$$

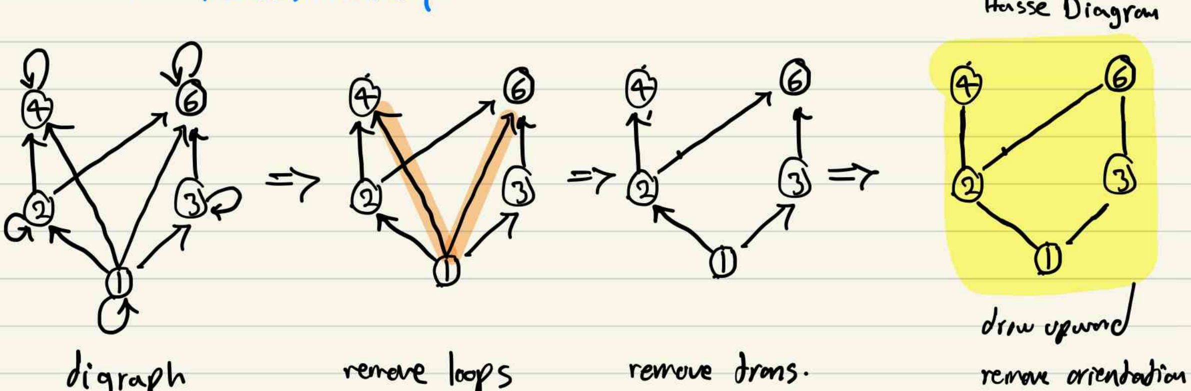
$$\left.\begin{array}{l} x R^{-1} y \text{ then } y R x \\ y R^{-1} z \text{ then } z R y \end{array}\right\} \Rightarrow z R x \text{ therefore } x R^{-1} z \; \checkmark$$

Hasse Diagram: a graphical representation of partial orders

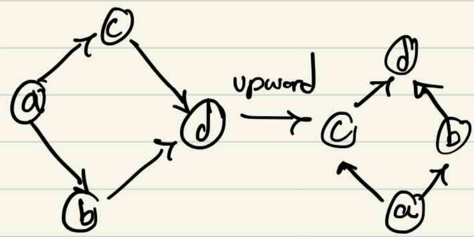
- start with digraph
- remove all loops (reflexive edges)
- remove all transitive edges
- Orient remaining edges upward, remove orientation.

exmp: $A = \{1, 2, 3, 4, 6\}$

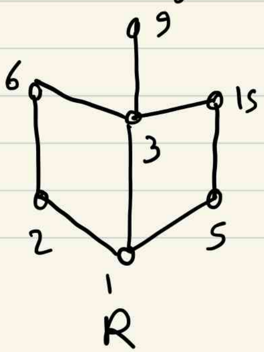
$R = \{(a,b) : a \mid b\}$



We can recover original relation from H-D.



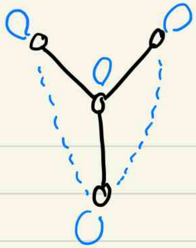
Give Hasse Diagram of R, write R as a set:



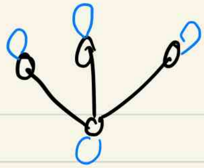
$R = \{(1,2), (1,3), (1,5), (2,6), (5,15), (3,6), (3,9), (3,15),$

$(1,1), (2,2), (3,3), (5,5), (6,6), (9,9), (15,15),$

$(1,6), (1,15), (1,9)\}$

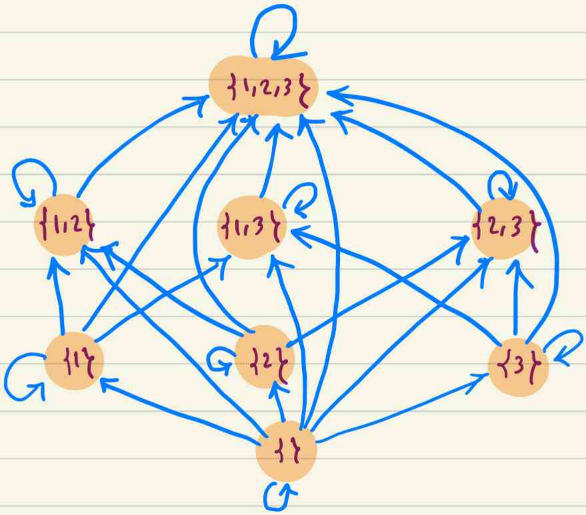


$|R| = 9$

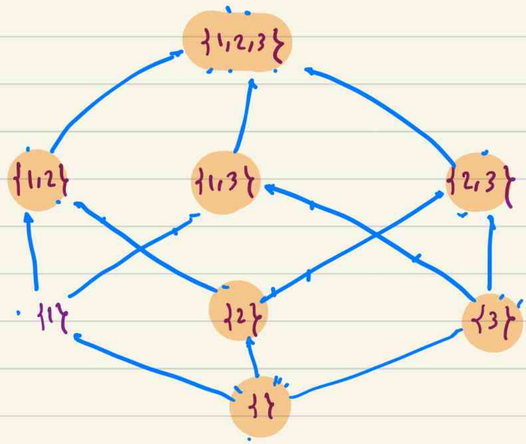


$|R| = 7$

exmp: $A = \{1,2,3\}$ and $\subseteq$ is a partial order on 2^A.



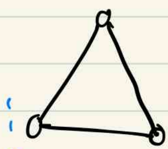
digraph



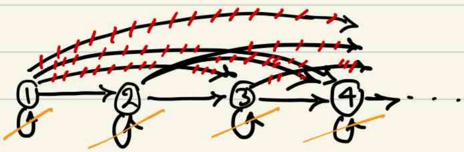
Hasse Diagram

In H.D. there is no cycle of length 3.

(one of edges must be reversed by transitivity)



$(\leq, \mathbb{N})$

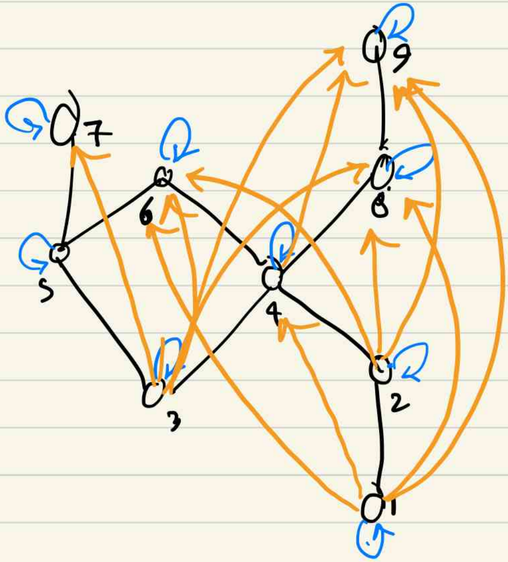


digraph

Hasse diagram

If the Hasse diagram is chain then the relation is called <u>total order</u>.

$(\leqq, \mathbb{N})$ is a total order.



$|R| = ?$

29

32

25

~~26~~

$9 + 9 + 12 = 30$